

WORK EXPERIENCE

SENIOR RESEARCHER - ROBOT LEARNING

Mar 2026 - present



Abu Dhabi (UAE) Technology Innovation Institute: Humanoids Group

Developed a real-time Human-Robot Interaction stack for humanoid robots: spoken conversation, event triggering, custom wake-word models, and scene perception, description and question answering, cutting latency 10x. Designed a speech-synchronized gesture server that blends and chains, in real time, kinesthetically taught gestures via Laplacian trajectory editing.

RESEARCHER - AI AND ROBOTICS

Apr 2024 - Dec 2025



Rome (IT) Fucina Nexus Foundation (Remote)

Developed software for a social robot for elderly people and children: natural conversation, detection of anomalies and notable events, and real-time alerts to a custom dashboard.

RESEARCHER - COGNITIVE ROBOTICS AND MANIPULATION

Oct 2021 - Feb 2024



Istanbul (TR) Boğaziçi University: Cognitive Learning and Robotics Laboratory

Enabled industrial and collaborative robots to learn manipulative actions from few-shot demonstrations, adapt and perform different primitives on real-life objects, assess the results to plan and achieve long-term goals.

RESEARCHER - VIRTUAL REALITY

Jun 2022 - Dec 2022



Istanbul (TR) Boğaziçi University Virtual Augmented Reality Laboratory

Integrated particle-based physics for fluids, cloth and deformable objects in Unity VR. Built and animated 3D assets via LiDAR scanning, photogrammetry, mesh cleanup and re-texturing.

RESEARCH ASSISTANT - INTERACTION ROBOTICS

Apr 2020 - Sep 2020



Stockholm (SE) KTH Royal Institute of Technology: Robotic Perception and Learning Department

Developed natural-language interaction software for a humanoid robot: dialogue engine, multi-turn conversation handling, and persistent per-user memory to personalize interactions.

RESEARCH SCHOLAR - COMPUTER VISION

Sep 2018 - Sep 2019



Lisbon (PT) IST Instituto Superior Técnico: I.S.R. Institute for Systems and Robotics - VisLab

Built the perception and navigation stack of a mobile robot: visual SLAM and autonomous indoor navigation, Bluetooth-beacon localization, face detection, object recognition, and an NLP module for voice interaction and parametrized task execution.

PUBLICATIONS

Novel Robotic Action Synthesis with Conditional Neural Movement Primitives.

Lirussi, I. *MSc thesis*, Computer Science and Engineering, University of Bologna, Dec 2023. Supervisor: Prof. Emre Ugur (Boğaziçi University) [PDF]

Framework for Navigation and Human-Robot Interaction with Humanoid Robots.

Lirussi, I. *BSc thesis*, Computer Science and Engineering, University of Bologna, Sep 2019. Supervisor: Prof. Alexandre Bernardino. [PDF]

EDUCATION



MSc. COMPUTER SCIENCE AND ENGINEERING, *University of Bologna* Sep 2019 - Dec 2023

Exchange MSc student at Stockholm University: Network Security, BI, and Cyber Forensics.



TECHNICAL SKILLS & SOFTWARE

Languages: Python, C++, C, C#, Java, Bash.

Robotics: ROS 2, MuJoCo, LfD, visual SLAM, MoCap, VR teleoperation.

AI: PyTorch, OpenCV, LLM/VLM and real-time speech APIs.

Tools: Docker, Linux, NVIDIA Jetson, FastAPI, GitLab CI, Jira.

 ROS / ROS 2
 PyTorch
 Docker
 OpenCV
 Git/GitLab/GitHub CI

LANGUAGES & INTERESTS

	IT	Native Speaker	Winner "The 30 new emerging ideas of 2021" - Startup Day Unibo
	EN	Full Proficiency	Theoretical and Practical course of First Aid for companies
	PT	Conversational	Cooking and DIY Enthusiast
	SV	Basic level	Sport and Videomaking Passionate
	TR	Basic level	Food Aid Program Volunteer - Blood Donor